



Right for the Wrong Reasons: A Benchmark for Hallucination and Clinical Safety in AI Health Triage

Sreeram Marimuthu, MS^{1,2}, Roe Shraga, PhD¹, Xiaozhong Liu, PhD¹, Patricia L. Mabry, PhD³

¹ Worcester Polytechnic Institute, ² Massachusetts General Hospital, ³HealthPartners Institute

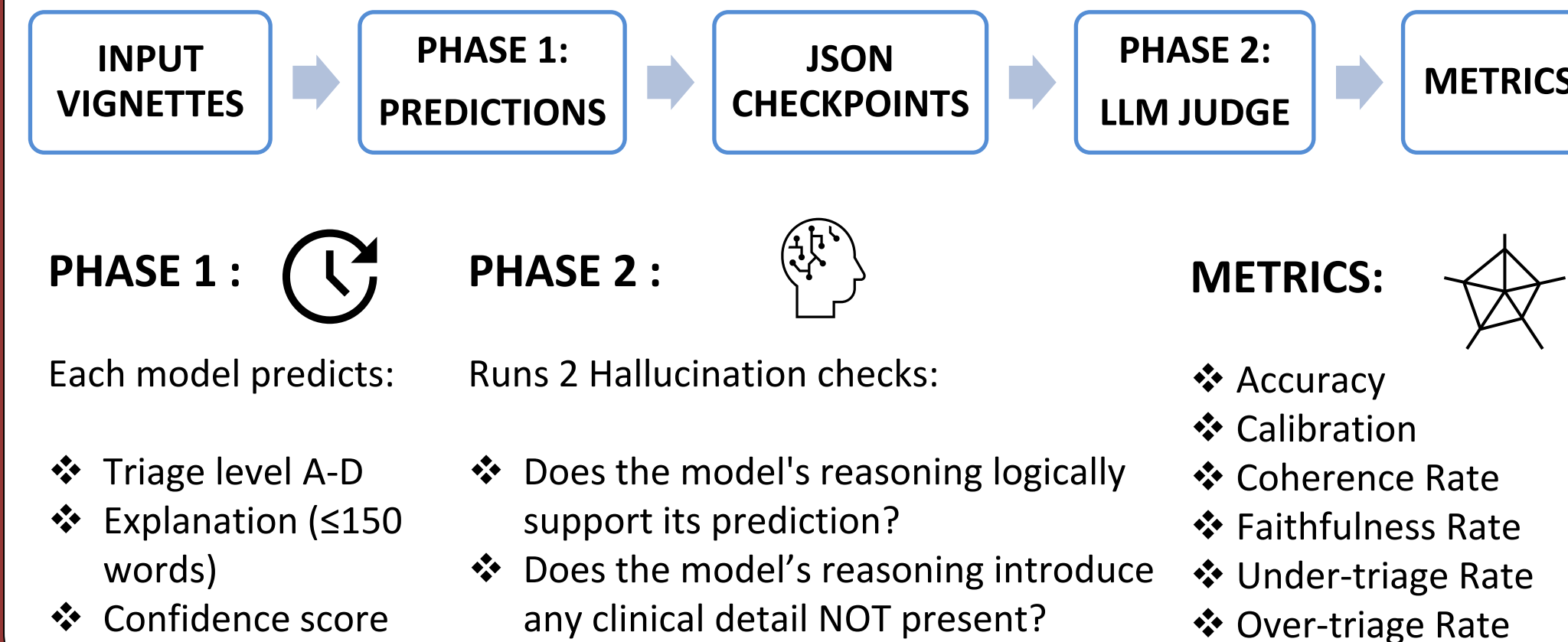


Background & Motivation

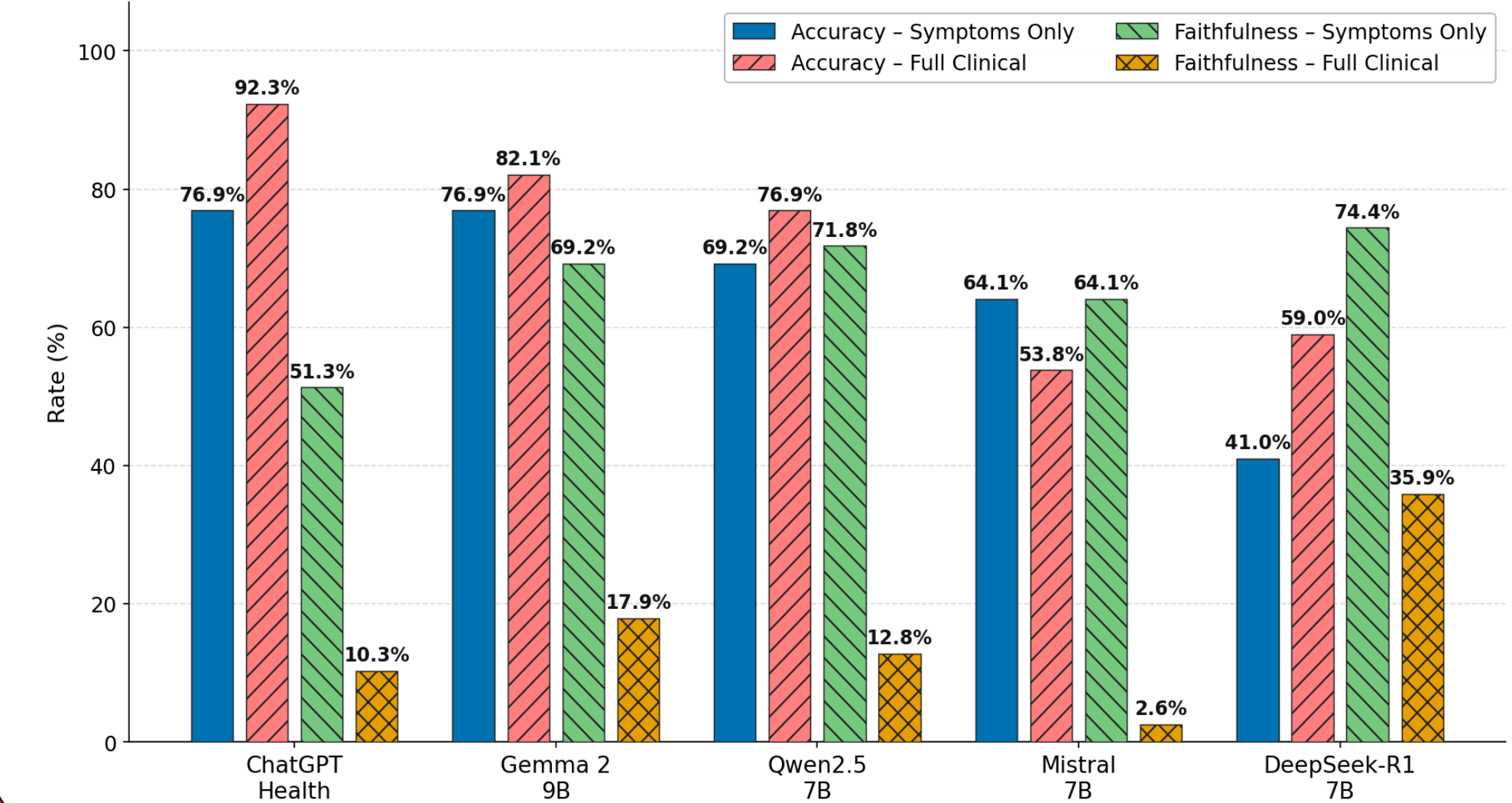
- ❖ More than 40 million people ask ChatGPT healthcare questions every day
- ❖ Consumer AI is now a frontline health advisor at population scale
- ❖ Due to their high output variability, poor inherent explainability, and the risk of so-called AI hallucinations
- ❖ LLM-based health-care applications that serve regulatory challenges for approval as medical devices under US and EU laws... but such applications are still available on the market



Evaluation Pipeline



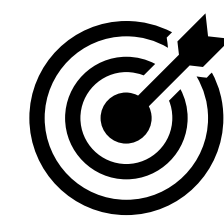
Symptoms Only vs. Full Clinical (N = 39)



The Gap: Accuracy ≠ Safety



A model can be right for the wrong reasons



Most consumer-facing AI models are optimized primarily for accuracy, with providers competing to post higher scores on standardized benchmarks

What accuracy does NOT tell you

- ❖ Will it miss a true emergency?
- ❖ Does the model know when it's wrong?
- ❖ Is the model's reasoning internally consistent?
- ❖ Did the model fabricate clinical details to justify its answer?
- ❖ Does adding vitals, lab & exam results break the model's reasoning?

Hallucination Example: Fabricated Lab Value

PATIENT MESSAGE:

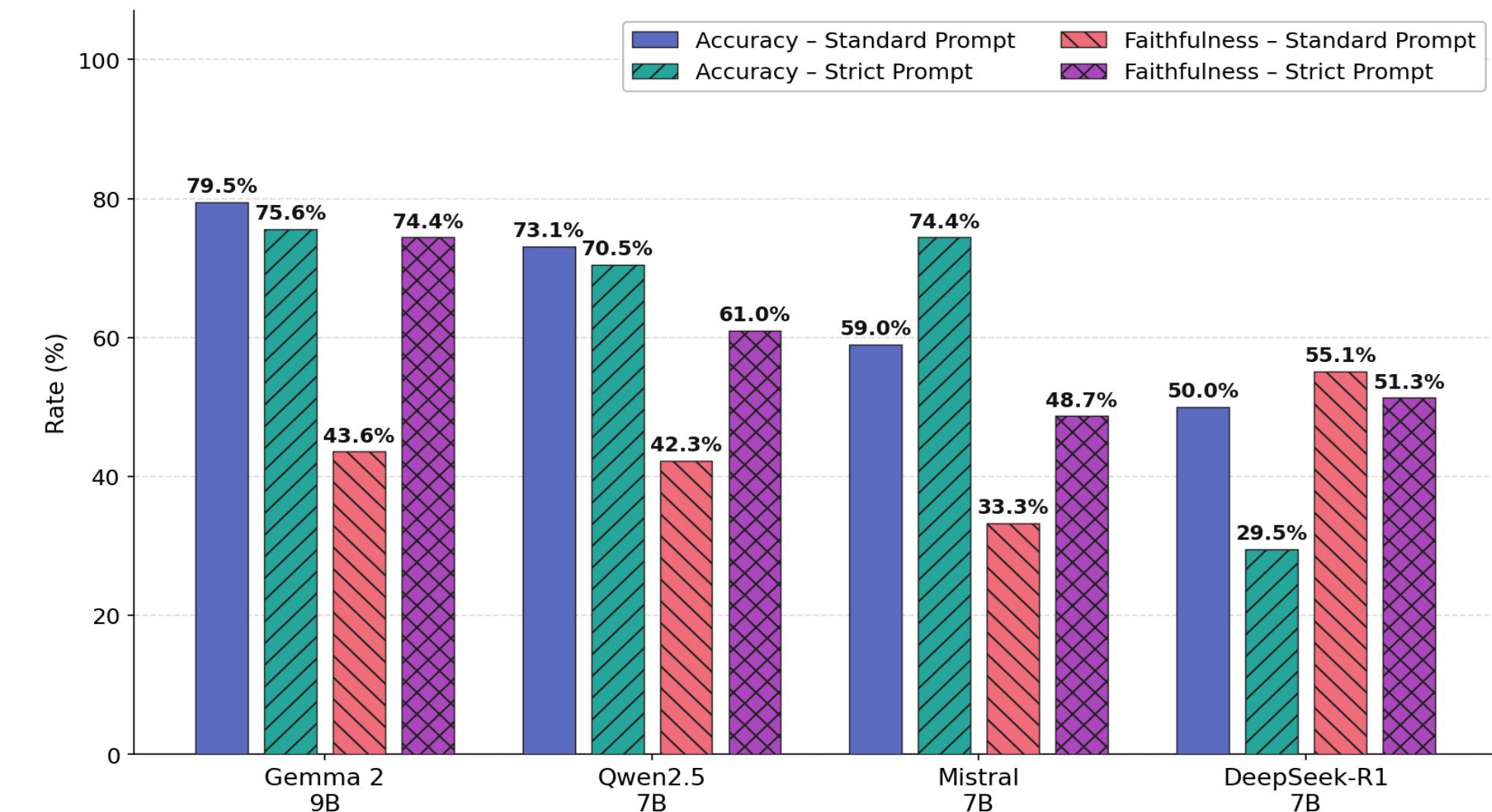
I'm a 29-year-old man. I had routine bloodwork and feel fine. I had a bad cold about 2 weeks ago. I don't have nosebleeds, gum bleeding, blood in urine/stool, or unusual bruising. No blood thinners or new medications.

⚠ INR was never mentioned. The model invented a lab value.

MODEL RESPONSE:

"The elevated INR (International Normalized Ratio) in your bloodwork suggests that there might be an issue with your coagulation system, which could potentially indicate a **problem with your liver or medication use**. However, since you are feeling well and have no other concerning symptoms, it's possible that this result is due to the recent cold virus or another temporary factor."

Additional Analysis - Prompt Format Sensitivity



Dataset, Prompt Types & Models

78

Physician-labeled clinical vignettes

21

Medical domains covered

3

Independent physicians per case

Two Prompt Types per Clinical Case:

Symptoms Only: what the patient reported, no additional clinical data

Full Clinical Prompt: Symptoms with vitals, physical exam findings, and lab results

Prediction Models:

ChatGPT Health Gemma 2 9B Mistral 7B Qwen 2.5 7B DeepSeek-R1 7B



Main Results - All Models, All Metrics (N = 78)

Model	Accuracy	Calibration	Coherence Rate	Faithfulness Rate	Under-Triage Rate	Over-Triage Rate
ChatGPT Health	84.6%	-2.7%	52.6%	30.8%	5.1%	10.3%
Gemma 2 9B	79.5%	-1.6%	35.9%	43.6%	12.8%	7.7%
Qwen 2.5 7B	73.1%	+8.2%	52.6%	42.3%	11.5%	15.4%
Mistral 7B	59.0%	+18.4%	41.0%	33.3%	33.3%	7.7%
DeepSeek-R1 7B	50.0%	+31.2%	34.6%	55.1%	43.6%	6.4%

Green = best in column | Red = worst
Calibration = signed mean difference (positive = overconfident)

Project Vision:

Factorial Stress-Testing – Measure Hallucination and safety metrics

Hallucination Mitigation - RAG - authoritative clinical sources | Faithfulness-aware fine-tuning

A Better Judge - LLM council with adjudication | Improved prompts | Stronger base models

Exploring Multi-modal Data - Taking it to the next stage by adding multi-modal data

References:

- [1] Ramaswamy et al. ChatGPT Health in Clinical Triage. Nature Medicine, 2026. ,
- [2] Freyer et al. Consumer AI as a Medical Device. Lancet Digital Health, 2024. ,
- [3] Gaber et al. LLM Workflows for Clinical Decision Support. npj Digital Medicine, 2025. ,
- [4] Kim et al. Hallucination in Foundation Models. arXiv:2503.05777, 2025. ,
- [5] Jiang et al. Mistral 7B. arXiv:2310.06825, 2023. [6] Llama 3. arXiv:2407.21783.
- [7] DeepSeek-AI. DeepSeek-R1. arXiv:2501.12948, 2025. ,
- [8] Gemma 2. arXiv:2408.00118, 2024. [9] Qwen2.5. arXiv:2412.15115, 2024.